

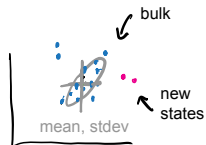
Classical Systems Biology
MCSB Bootcamp — Mathematical and Computational Track

Jun Allard

The goals, challenges, systems, tools and approaches of today

Data and analytics

Keystone **goal**: discover the hidden — new states, cell types, disease classes, ...



Keystone **challenge**: biological data is not just big, it is **high-dimensional**.

- ▶ 30000 genes, 1000 cells
- ▶ 10^4 microbial species, 20 patients



Keystone **tool**: clustering, regression, dimension-reduction ... coding.

Models and mechanism

Keystone **goal**: **prediction** — “what would happen if...”
if **A** is increased, then...



Keystone **challenge**: biological systems are **complex**, not just big.



not complex



not complex

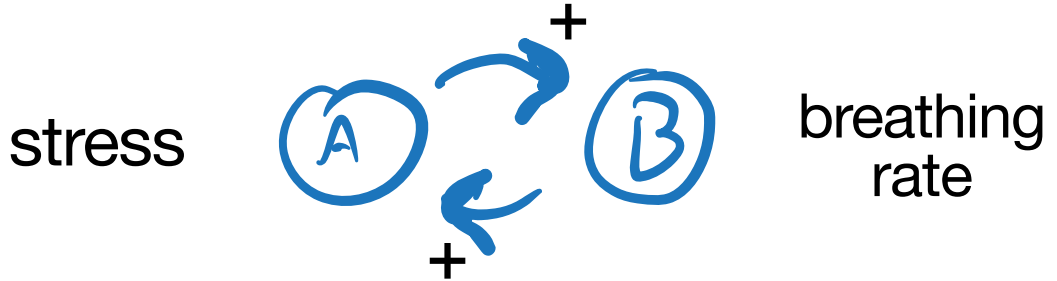


complex!



Keystone **tool**: dynamical systems, control theory, (the differential equation,) ...

Stress and breathing rate

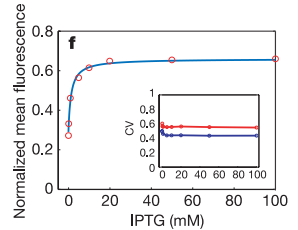


Complex systems and “systems thinking”

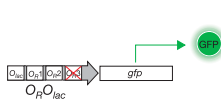
LETTERS

A bottom-up approach to gene regulation

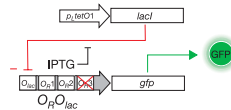
Nicholas J. Guido^{1*}, Xiao Wang^{2*}, David Adalsteinsson³, David McMillen⁵, Jeff Hasty⁶, Charles R. Cantor¹, Timothy C. Elston⁴ & J. J. Collins¹



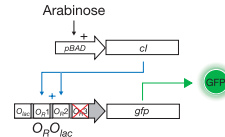
a Unregulated system



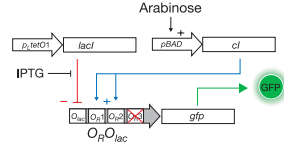
b Repressor-only system



c Activator-only system



d Repressor-activator system



“Computation drove the field of synthetic biology in its very early days.”

Collins, 2020

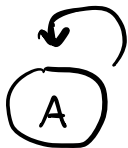
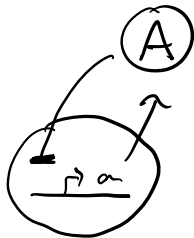
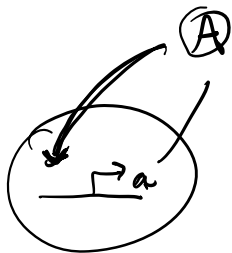
“We didn’t have those data sets. We didn’t have a wet lab. We didn’t have the capability. We had to rely on mathematical modeling.”

paraphrasing Collins, 2015

Complex biological systems and “systems thinking”

→ Activation

⊥ Repression



Positive feedback

With positive feedback, the curve showing steady-state protein as a function of activator TF should move up.

Does the curve move up by:

- ▶ shifting the EC50 to the left?
- ▶ moving up at high-C?
- ▶ moving up at low-C?
- ▶ a combination of all three?

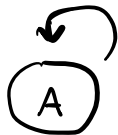
Seven circuits



Activation



Repression



Positive feedback

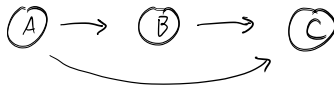
1



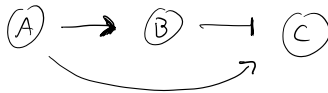
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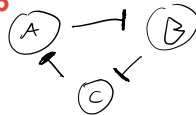
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4



5



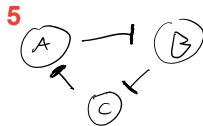
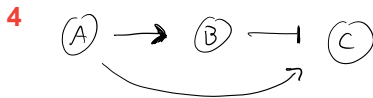
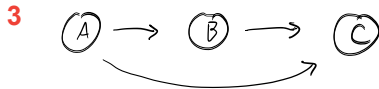
6



7



In-class activity: matching



Match each circuit to its name.

- ▶ Direct negative feedback
- ▶ Indirect negative feedback
- ▶ Double-negative feedback
- ▶ Fast positive and slow negative feedback
- ▶ Incoherent feedforward
- ▶ Coherent feedforward
- ▶ The Repressilator